Similarity by Selected Traits

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Similarity by Selected Traits

> Prepare Similarity by Selected Traits: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Represent items with numeric features

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Look for: Identify selection and coverage bias

Reference example, not the only acceptable wording: Afternoon or evening visitors, and people unable to attend in the morning.

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Look for: Detect absent and unusable values

Reference example, not the only acceptable wording: 0 is a recorded allowed duration. The other two are unknown under this schema.

Choose Informative Questions

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

Concept to explain

Week 21: **Similarity** by Selected Traits

Term	Meaning
similarity	How alike two items are according to selected features and a declared comparison method.
distance	A specified measure of how far apart two representations are.
coordinate	A value identifying position along an agreed axis or within a space.
scale	The units and range used to represent or compare a quantity.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 22 as a two-session investigation using fictional data. A fictional activity chooser represents each activity with energy and group-work ratings on a 0 to 5 **scale**. A learner requests [2, 4].

We compare options by adding absolute differences. This is a chosen rule for **similarity**; it is not a **learned model** or an objective measure of quality.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

The absolute difference is 4.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can calculate a feature **distance** and state how scaling changes its meaning.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Similarity by Selected Traits using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Reader target: length=2, difficulty=3.

2. Book A: (2,2) → grid **distance** |2−2|+|2−3|=1.

3. Book B: (4,3) → **distance** |4−2|+|3−3|=2.

4. A is closer under these features; changing features can change the answer.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

The teaching case

Use number of differing traits as **distance**.

Exact explanation to model

Why the result follows

Distances: A=1, B=0, C=1. B is nearest with equal feature weights.

Explain the added worked example

Explain the added worked example · 1

A fictional activity chooser represents each activity with energy and group-work ratings on a 0 to 5 **scale**. A learner requests [2, 4]. We compare options by adding absolute differences.

This is a chosen rule for **similarity**; it is not a **learned model** or an objective measure of quality.

abs gives a nonnegative gap. Under this rule Sketch is closer. Feature order must match: first energy, then group work. Both **scale**s are 0 to 5, but that does not prove one unit has identical practical meaning in both features.

Key terms

Term	Meaning
Feature vector	An ordered list of numeric features
Distance	A number describing difference under a chosen rule
Manhattan distance	The sum of absolute feature differences
Normalization	Rescaling features to a comparable scale

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **Find the distance from target [2, 4] to [2, 4] and to [3, 3].**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Similarity by Selected Traits

A fictional toy book matcher represents two preferences with numeric features.

The facts we will use

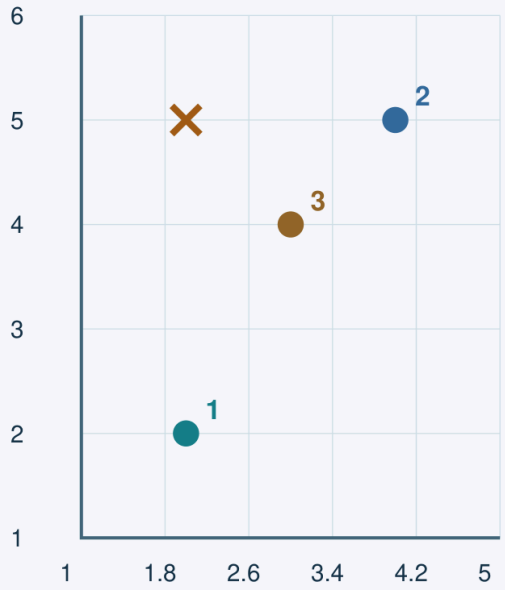
> **Supplied facts**
>
> Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Supplied facts

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

AI ACADEMY / AI-2/21

Toy feature positions



Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/21

Similarity by Selected Traits

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Distance A

$|2-2|+|5-2|=0+3=3$.

2. Distance B

$|2-4|+|5-5|=2+0=2$.

3. Distance C

$|2-3|+|5-4|=1+1=2$.

4. Choose closest

B and C tie at distance 2; keep the tie visible.

CHANGE ONE THING

Give the first-coordinate difference weight 2 while the second keeps weight 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Distance A	$\lvert 2-2\rvert+\lvert 5-2\rvert=0+3=3$.
2. Distance B	$\lvert 2-4\rvert+\lvert 5-5\rvert=2+0=2$.
3. Distance C	$\lvert 2-3\rvert+\lvert 5-4\rvert=1+1=2$.
4. Choose closest	B and C tie at distance 2; keep the tie visible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Each **coordinate** contributes according to the declared **distance** rule. Smaller **distance** means closer in this representation, not automatically a better reading experience.

If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

What this example does not establish

Keep the stated limit

A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

Change one thing in the pictured case

Change one thing

Change: Give the first-**coordinate** difference weight 2 while the second keeps weight 1.

Predicted result

Prediction: **Distance**s become A=3, B=4, C=3; A and C tie.

Why the result follows

Reason: The representation’s comparison rule changes; the supplied vectors stay the same.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Similarity by Selected Traits, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

2. State the decision to explain

Write the question to explain: Use number of differing traits as **distance**.

3. Reveal the worked reasoning

Reveal after **prediction**s: **Distance**s: A=1, B=0, C=1. B is nearest with equal feature weights.

Hold back the independent task

Keep for an independent attempt: Score **similarity** using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Distances 0 and 2. Zero means identical recorded features, not identical activities in every way. Exit: No. Only the recorded features match; unrecorded needs and activity details may differ.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: From target [1, 2], find **distance**s to A [2, 2] and B [0, 4].

Reasoning to listen for

Teacher check: A: 1 + 0 = 1; B: 1 + 2 = 3.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Distance** between **feature vector**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: **Similarity** depends on the features collected and the **distance** rule. Missing accessibility, cost or preference information can make the closest item unsuitable.

Hard requirements should be checked separately.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Explain why B ceases to be a closest option without any candidate data changing.

Reasoned teacher answer

Its first-**coordinate** gap 2 now contributes 4, raising its **distance** above A and C's 3.

Changed-case reasoning: **Distance**s become A=3, B=4, C=3; A and C tie. The representation's comparison rule changes; the supplied vectors stay the same.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Similarity by Selected Traits investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Plot six books on a 2-feature grid, compute simple grid **distance**, find nearest, then change one selected feature and compare.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: **Similarity** depends on the features collected and the **distance** rule. Missing accessibility, cost or preference information can make the closest item unsuitable.

Hard requirements should be checked separately.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: target [2, 3]; X [0, 3], Y [2, 2], Z [3, 4] have **distance**s 2, 1, 2. Y wins and X/Z tie.

A convenient shortcut preserves meaning: If one feature is duration in minutes up to 60 and another a rating up to 5, raw **distance** may be dominated by duration.

Specify scaling or weights before ranking; do not silently change units to get a preferred winner.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Similarity by Selected Traits questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Distances: A=1, B=0, C=1. B is nearest with equal feature weights.

Independent question

Independent question · withhold the key

Score **similarity** using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

Expected independent answer

Teacher key · Expected independent answer

Each scores one point. The chosen traits create a tie; use a stated tie rule or request another relevant preference.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “**Similarity** always gives exactly one winner.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Equal-weight **distance**s are A=1, B=1, C=2. Doubling the illustration mismatch makes A=1, B=2, C=2, so A becomes nearest. Weight choices express a task preference; they are not objective truth about book quality.

Explanations for the additional workbook examples

W1 Read a vector

Workbook question · W1

For [4, 1] under the worked feature order, identify energy and group-work ratings.

Teacher reasoning · W1

Energy 4, group work 1.

Teacher reasoning · W1

Use the declared feature order: position 0 means energy and position 1 means group work. The vector [4,1] therefore records energy 4 and group work 1.

Reversing the interpretation would change the meaning and could change the **distance** calculation.

W2 Compute distances

Workbook question · W2

From target [1, 2], find **distance**s to A [2, 2] and B [0, 4].

Teacher reasoning · W2

A: 1 + 0 = 1; B: 1 + 2 = 3.

W3 Repair feature order

Workbook question · W3

One record uses [group work, energy] while others use [energy, group work]. Why is the **distance** unreliable?

Teacher reasoning · W3

It compares different meanings in each position. Reorder using the documented schema before computing.

W4 Rank independently

Workbook question · W4

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate **distance**s, select the closest, and identify a tie among the others.

Teacher reasoning · W4

Distances A 2, B 2, C 1. C is closest; A and B tie. Do not claim a unique ranking of the tied pair without a rule.

W5 Limit similarity

Workbook question · W5

Two activities have **distance** zero. Must they suit every learner equally well?

Teacher reasoning · W5

No. Only the recorded features match; unrecorded needs and activity details may differ.

Assess independent transfer

Independent question · withhold the key

Selected case: Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate **distance**s, select the closest, and identify a tie among the others.

Selected case answer

Teacher key · Selected case answer

Distances A 2, B 2, C 1. C is closest; A and B tie. Do not claim a unique ranking of the tied pair without a rule.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Similarity by Selected Traits to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Make selected traits and their **scale**s visible in every recommendation explanation.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 22: Nearest-Neighbor Reasoning. Retrieve today’s **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use a small square grid and count horizontal plus vertical steps. Extension: re**scale** a minutes feature by its declared range and compare rankings transparently.

Independent W4 **evidence**: **Distance**s A 2, B 2, C 1. C is closest; A and B tie. Do not claim a unique ranking of the tied pair without a rule. Record whether the learner needed no help, a hint or explanation.

Fresh retry: target [2, 3]; X [0, 3], Y [2, 2], Z [3, 4] have **distance**s 2, 1, 2. Y wins and X/Z tie.

Next week: Weighted scores and constraints. Reuse this week's explicit contracts and **evidence**.

Plan the delayed return

Suggested return points in this level: ai-2/22, ai-2/24, ai-2/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	

Record	Teacher evidence
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Similarity by Selected Traits lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Represent items with numeric features

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: From target [1, 2], find **distance**s to A [2, 2] and B [0, 4].

> ### Supported practice answer
>
> Reasoned practice answer: A: 1 + 0 = 1; B: 1 + 2 = 3.

Check every learner before discussion: Use the worked case for **Distance** between **feature vector**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: **Similarity** depends on the features collected and the **distance** rule. Missing accessibility, cost or preference information can make the closest item unsuitable.

Hard requirements should be checked separately.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Require a **prediction** and a reason before comparing. Expected result: **Distance**s become A=3, B=4, C=3; A and C tie.

Reason: The representation’s comparison rule changes; the supplied vectors stay the same.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately

Criterion	Not yet	With support	Independent
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

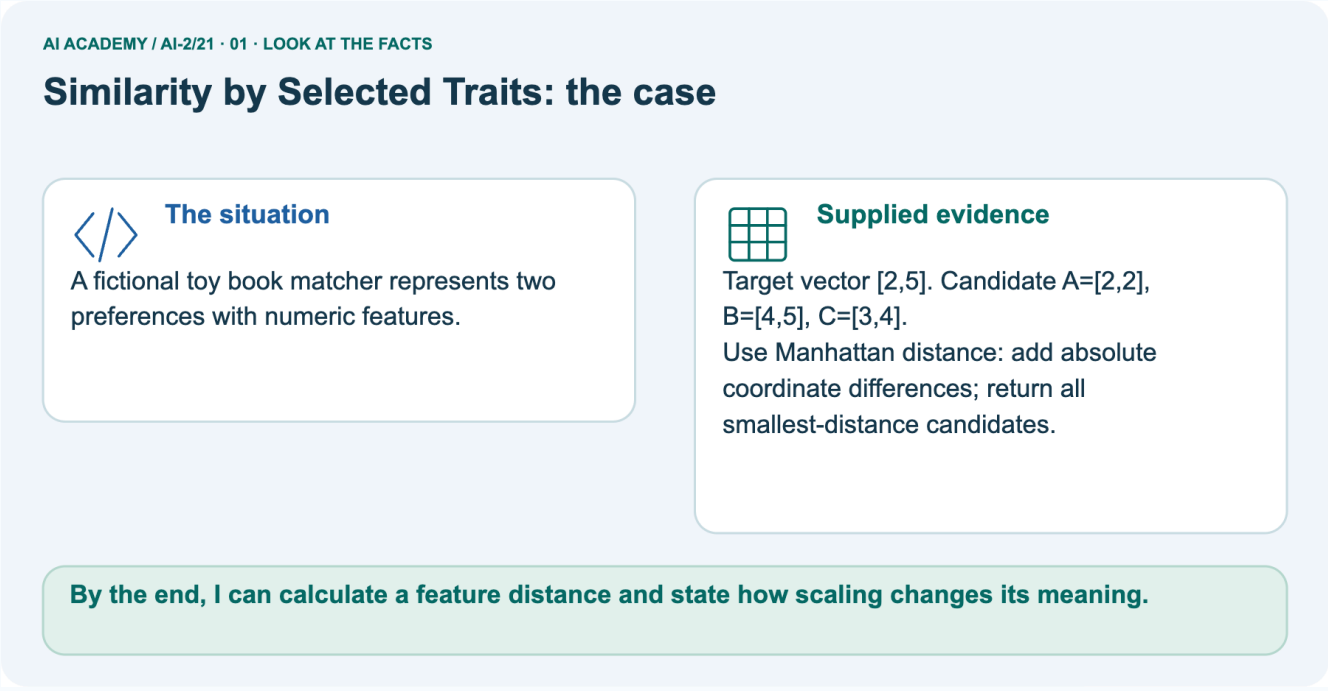
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Visual concept journey

> Show one Similarity by Selected Traits graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Similarity by Selected Traits: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional toy book matcher represents two preferences with numeric features. Supplied facts: Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

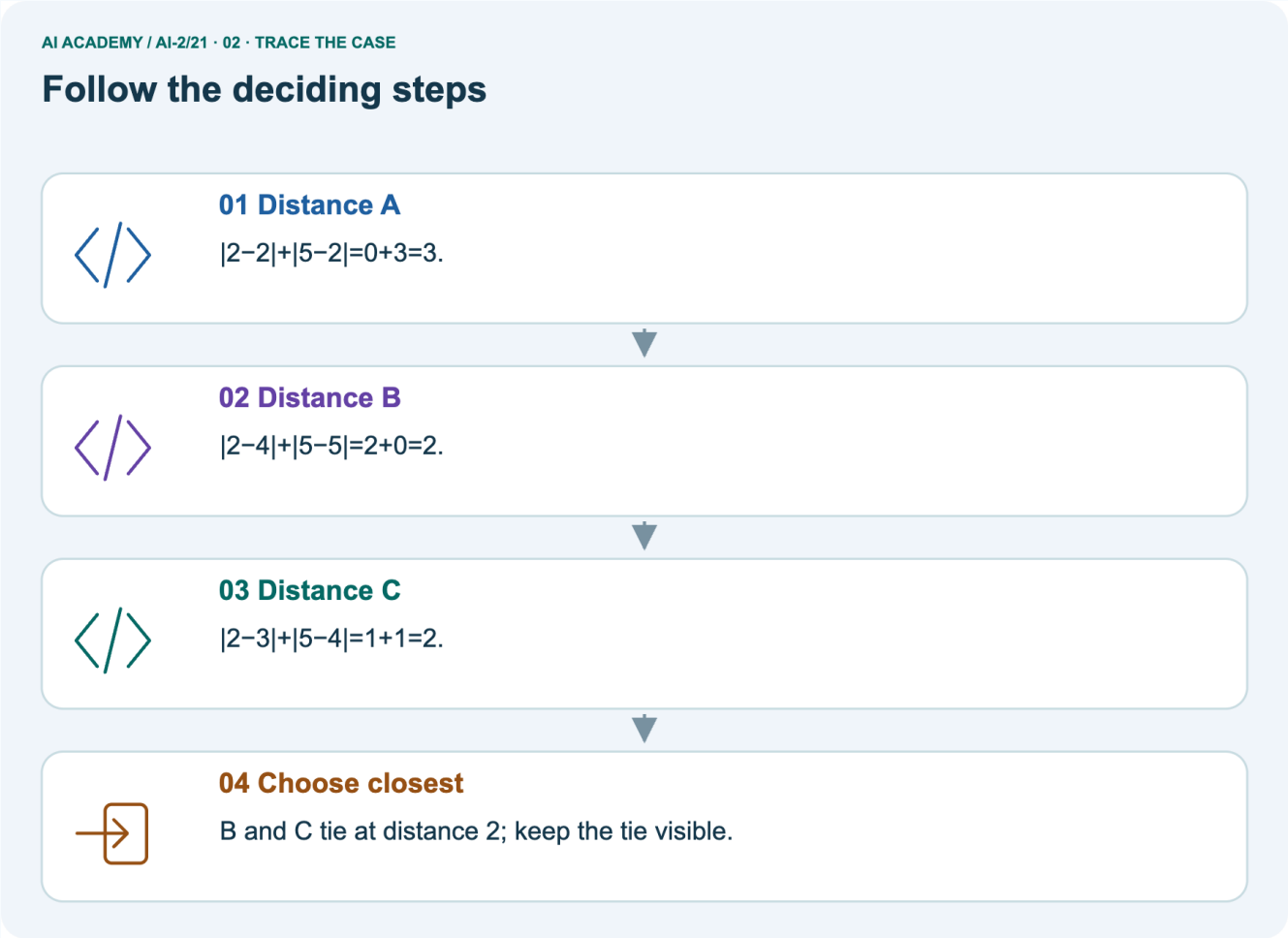
Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Distance A: |2−2|+|5−2|=0+3=3. → **Distance** B: |2−4|+|5−5|=2+0=2. → **Distance** C: |2−3|+|5−4|=1+1=2. → Choose closest: B and C tie at **distance** 2; keep the tie visible.

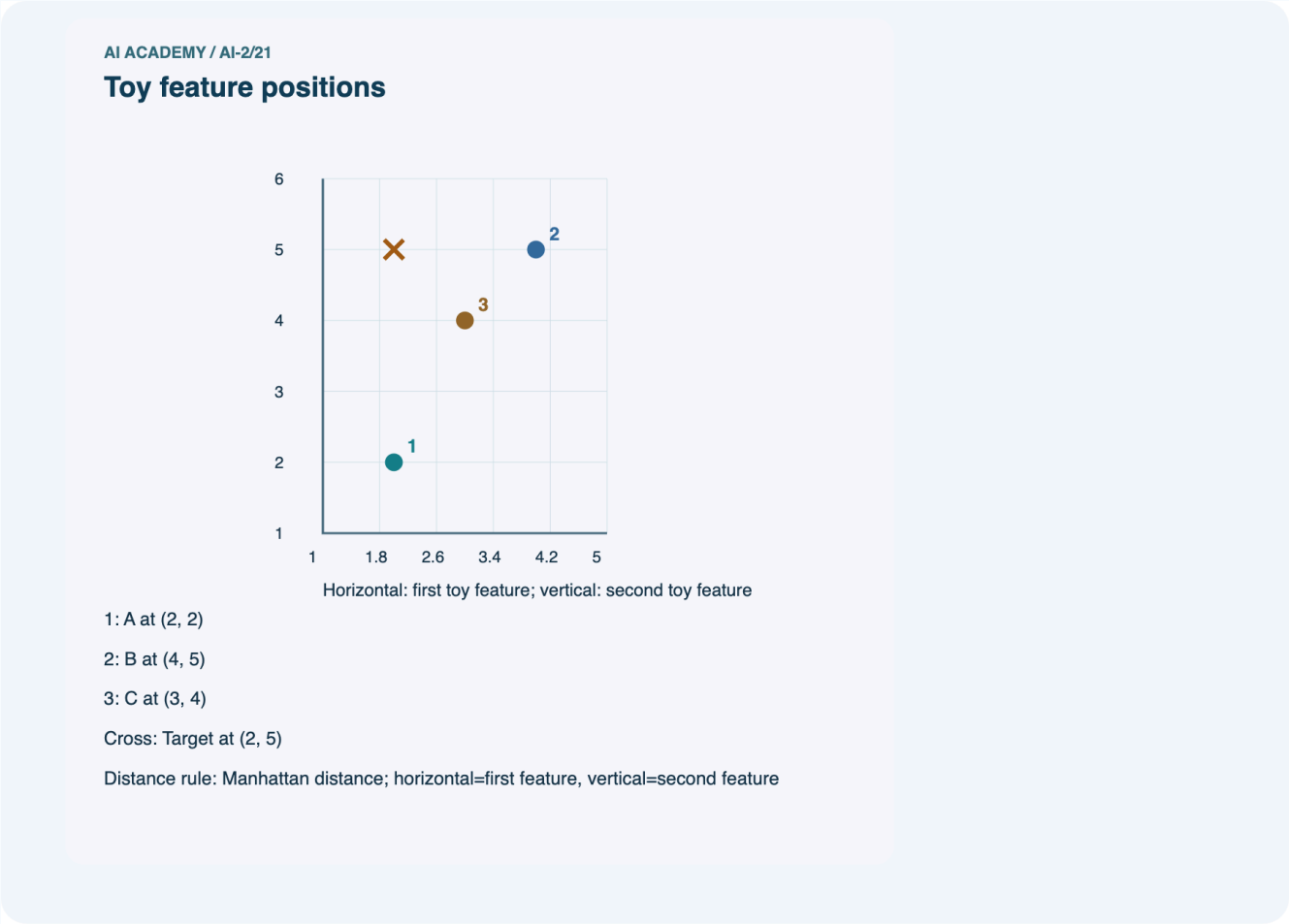
Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if

the image is difficult to see.

03. Toy feature positions



Follow the labelled picture with the matching explanation. Open the vector diagram

Toy feature positions. A: (2, 2). B: (4, 5). C: (3, 4). Query Target: (2, 5). Axes: first toy feature; second toy feature. **Distance**: **Manhattan distance**; horizontal=first feature, vertical=second feature.

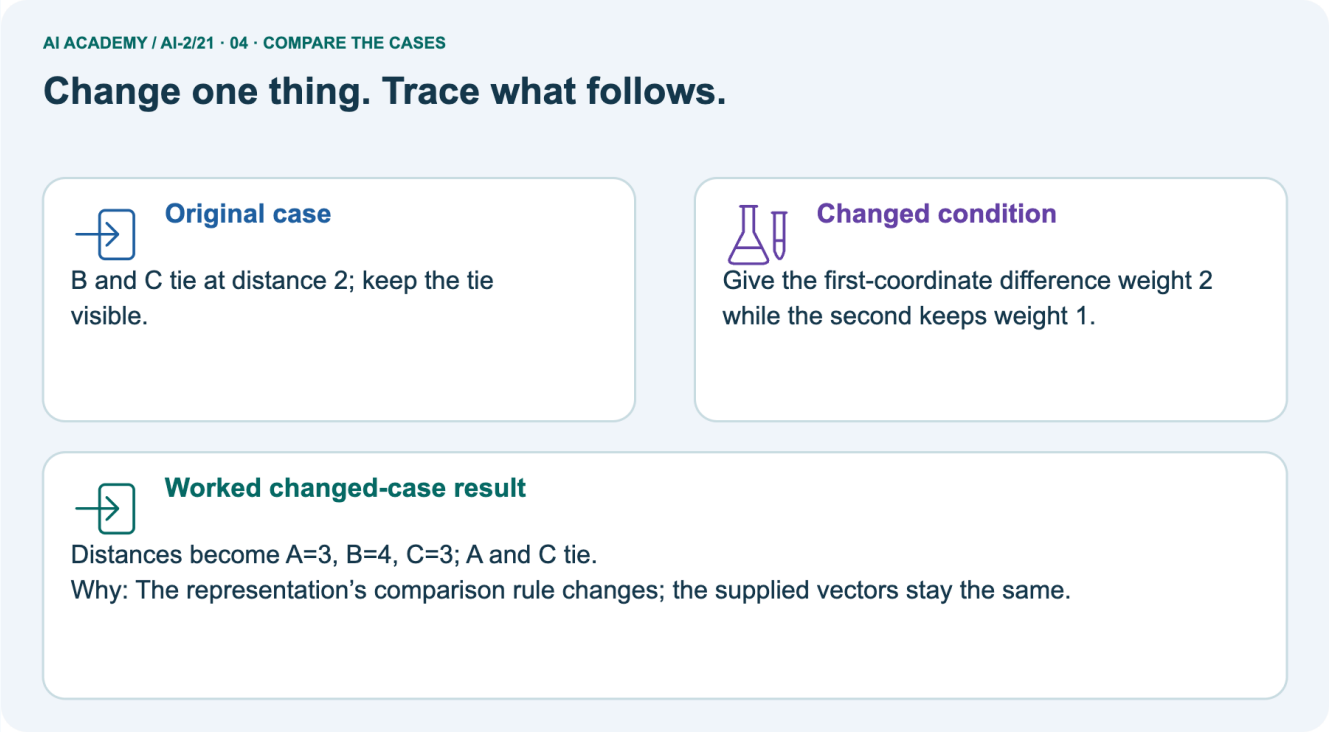
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B and C tie at **distance** 2; keep the tie visible. Changed condition: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Expected result: **Distance**s become A=3, B=4, C=3; A and C tie.

Reason: The representation's comparison rule changes; the supplied vectors stay the same.

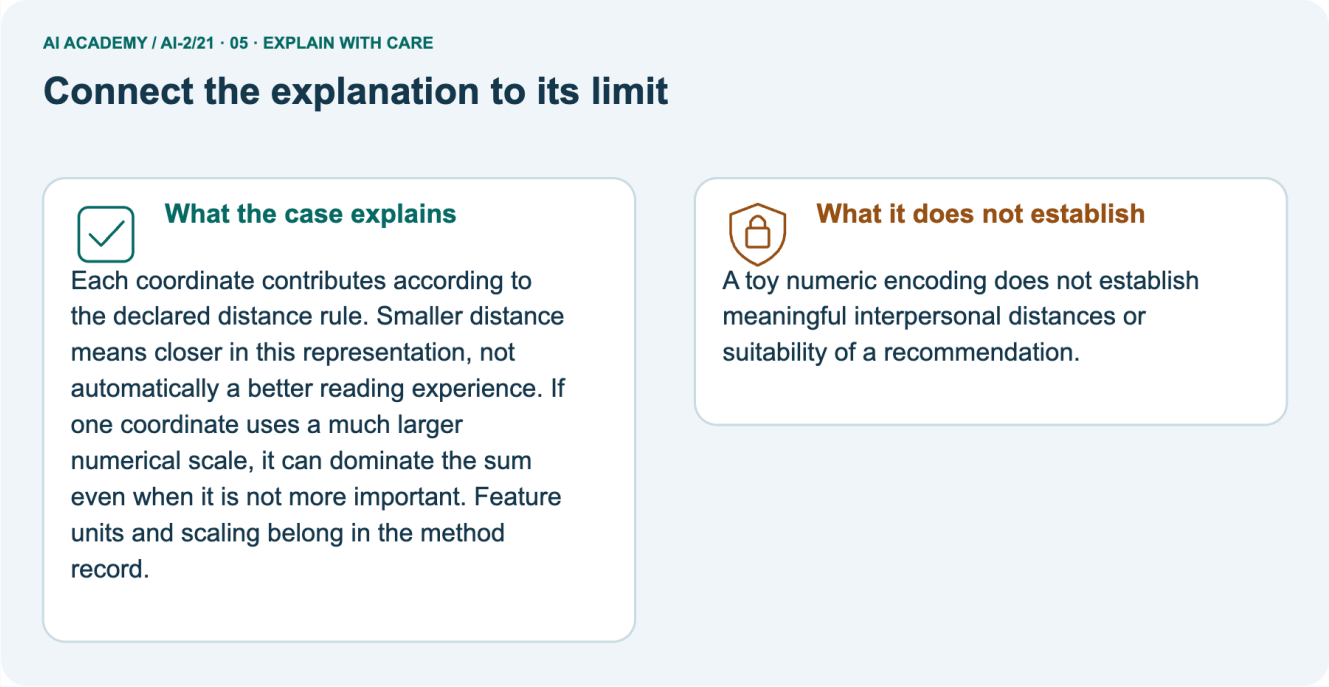
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The representation’s comparison rule changes; the supplied vectors stay the same.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Each **coordinate** contributes according to the declared **distance** rule. Smaller **distance** means closer in this representation, not automatically a better reading experience.

If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

Limit: A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

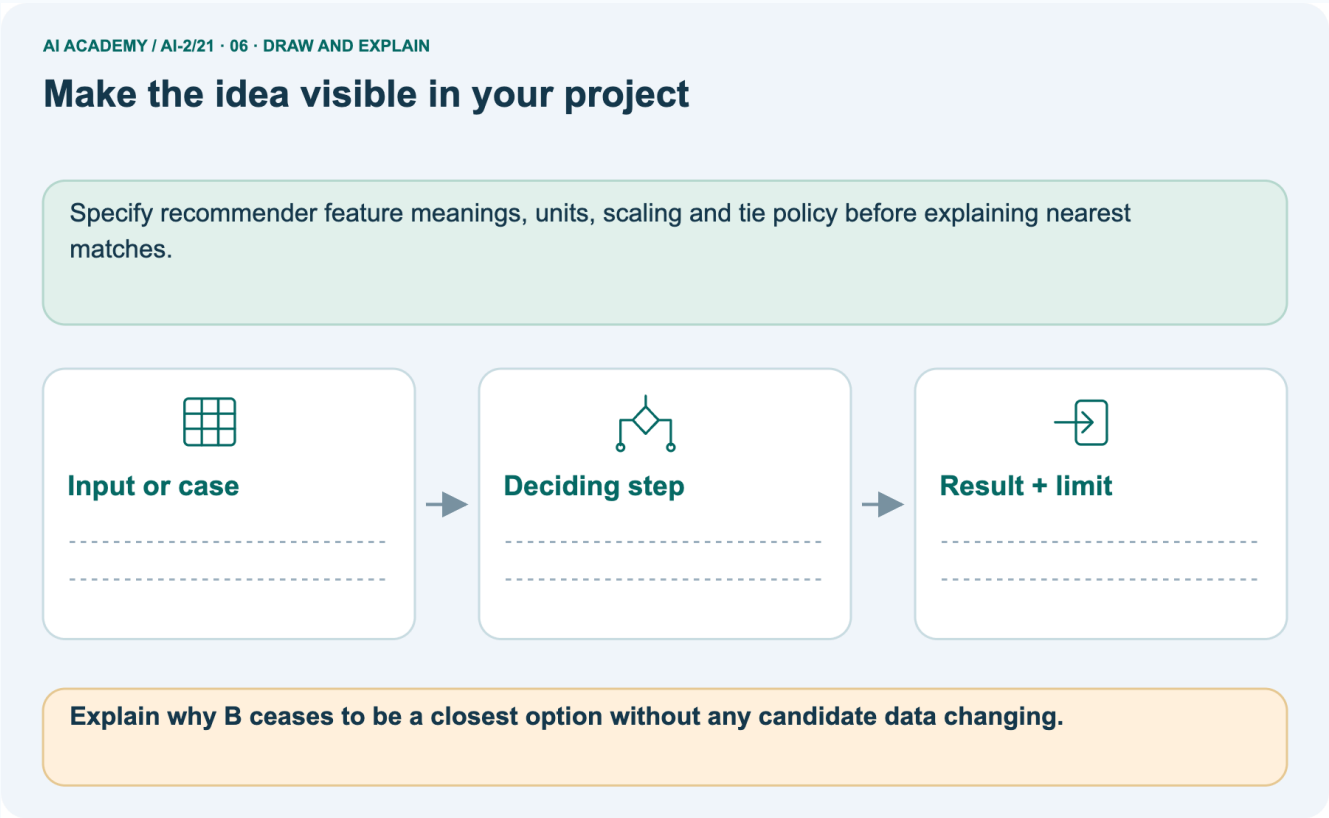
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B ceases to be a closest option without any candidate data changing.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: Its first-**coordinate** gap 2 now contributes 4, raising its **distance** above A and C’s 3.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.